# Forward Reasoning in Isabelle (FW)

Lecture Notes

**Course:** Computer-Assisted Theorem Proving (7CCSACTP)

## 1. Context & Overview

→ **Last Video Revision:**

- **Basic Functional Programming:** Currying, Lambda expressions, Polymorphic types.
- **Isabelle Interface:** Theory files, terms (numbers, lists, sets, products/tuples, functions) and type annotations.
- **Logical Specifications:** Connectives and theorem structure.

→ **Performing Proofs in Isabelle:**

- **Logical Reasoning:** Forward reasoning (FW) & Backward reasoning (BW).
- **Proofs by Substitution.**
- **Mathematical Induction** on natural numbers (`nat`).

## 2. Forward Reasoning Mechanics: Example Proof

**Goal Statement:** $P \to Q, \quad Q \to R, \quad \neg R \vdash \neg P$

**Step-by-Step Derivation:**

1. $\neg R$ *(Assumption 1)*
2. $Q \to R$ *(Assumption 2)*
3. $Q \to \neg R$ *(from 1, using $\to I$ rule)*
4. $\neg Q$ *(from 2, 3, using $\neg I$ rule)*
5. $P \to \neg Q$ *(from 4, using $\to I$ rule)*
6. $\neg P$ *(from assumption $P \to Q$ and 5, using $\neg I$)* ✓

**Inference Rules & Isabelle Syntax:**

- Implication Introduction ($\to I$):

$$\frac{X}{Y \Rightarrow X}$$

- Negation Introduction ($\neg I$):

$$\frac{X \to Y \qquad X \to \neg Y}{\neg X}$$

- **Isabelle Forward Attribute (`OF`):**

$$f_2 \ \texttt{[OF } f_1\texttt{]}$$

Instantiates premise of rule $f_2$ with known fact $f_1$ (e.g. `impI[OF fact1]`).

## 3. Isabelle Forward Reasoning Notes

- **Using Facts:** To obtain and utilize a fact $P$ in Isar:

```
have P    using P
```

- **Schematic Variables:** In Isabelle, schematic variables written with a question mark (e.g. `?x`) represent universally quantified variables ($\forall x$).